GEOM 3007 - Final Report

Theme C: Magic Mushroom Store

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Pro Map

Statement of Scope:

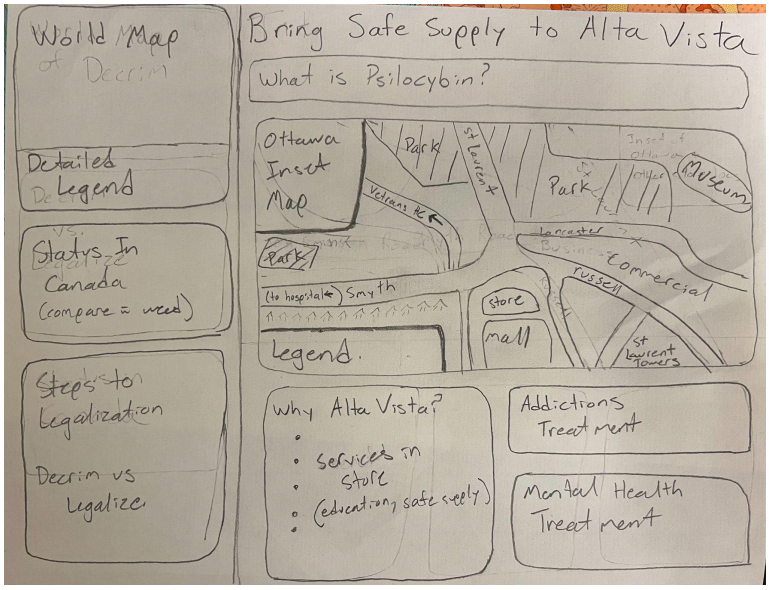
The purpose of this map is to demonstrate to the city counselor of Alta Vista (Ward 18), Marty Carr, why it would be advantageous to open a magic mushroom ('shroom') store in the Elmvale Acres Shopping Centre, located at 1910 St. Laurent Boulevard, Ottawa, Ontario. This map will communicate the benefits that would be brought to the Alta Vista community by a new magic mushroom store that is focused on mental health and addiction treatment. Benefits include education about safe use, access to a safe supply, and self-treatment options based on recent research; information included will be based on academic and grey literature. Map will be sized to fit on a projector screen and will be accompanied by a presentation to Counselor Carr, as well as an 8.5x11 handout with additional information, to gain approval to move forward on the creation of this store in the Elmvale Acres Shopping Centre.

Design Description:

The 'pro' map will be a projector-sized display including three maps at small, medium, and large scales. The large scale map will show the immediate area surrounding the Elmvale Acres Shopping Centre with relevant nearby amenities and the location of our store highlighted. This will be accompanied by a medium scale, inset map of the broader Ottawa area showing the locations of other 'shroom' shops in the city. The small scale map will show an overview of psychedelic drug laws in other countries and jurisdictions. A panel on the side of the map will include information about the potential benefits of magic mushroom use and the reasons that the store is in alignment with City of Ottawa priorities; this information will be expanded on in the accompanying handout. Data sources include amenity information from Open Street Maps/Google Maps, Open Ottawa, and OpenDataSoft, with a custom basemap made with Mapbox Studio.

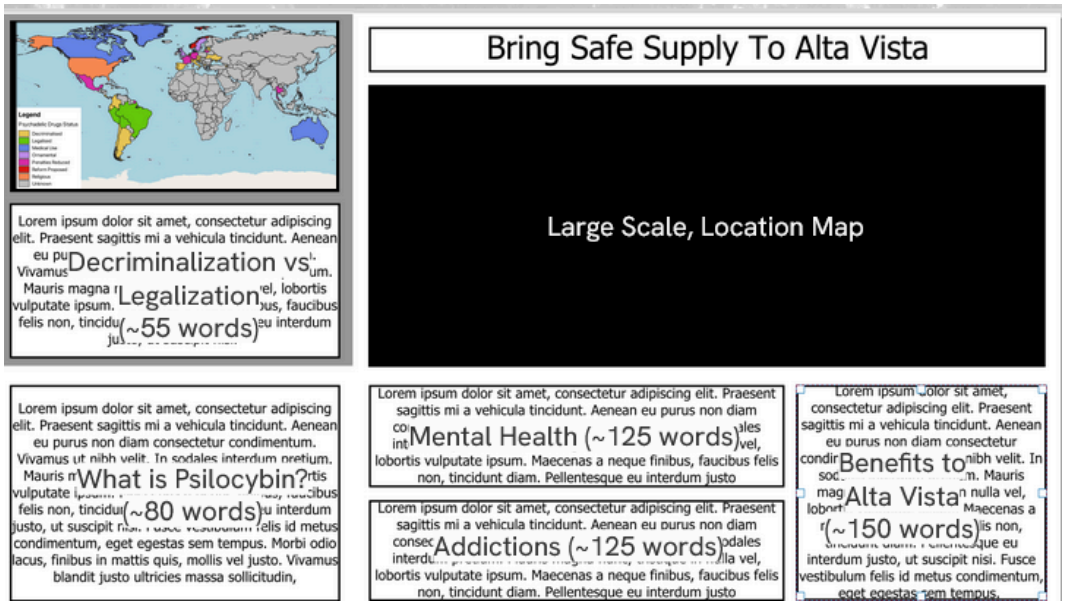
Map Drafts/Prototypes:

Initial Low-Fidelity Prototype (Feb 16th)



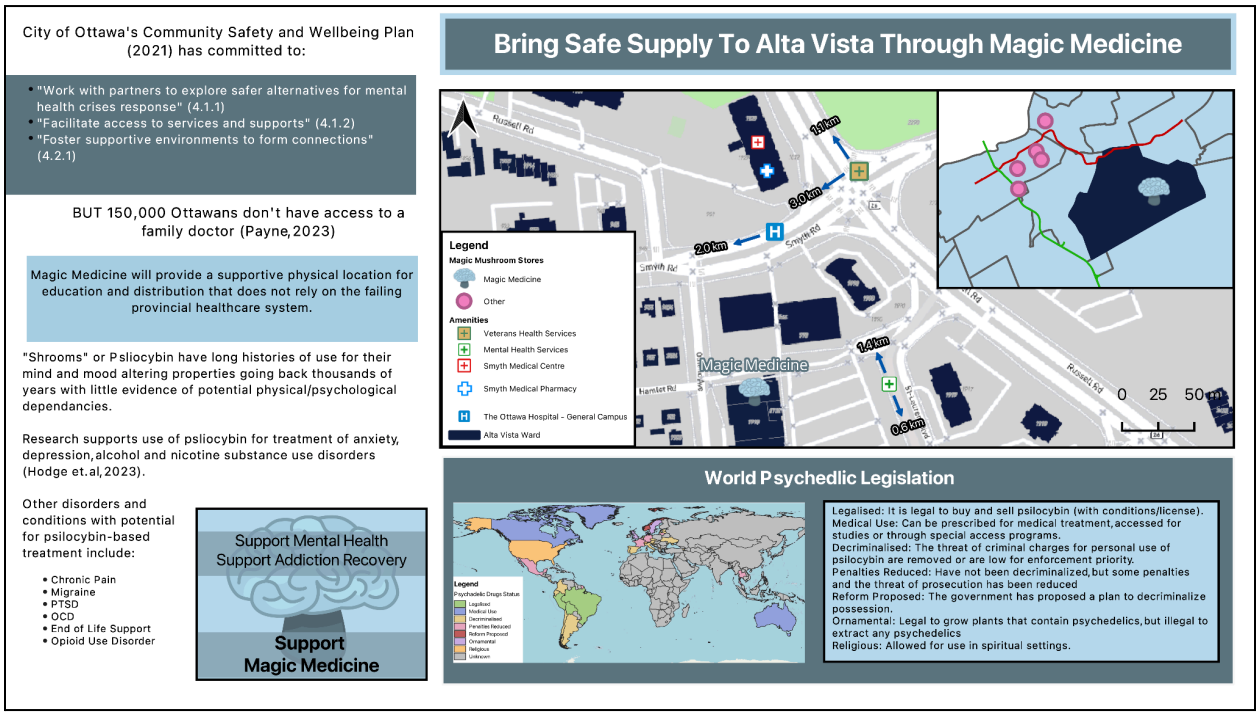
This image represents the initial low-fidelity mockup that was drawn on paper to represent our general goals; many of the aspects of this vision were kept or changed slightly as the process proceeded. Ottawa inset map was initially going to show overdose and neighborhood financial information, however this was later removed as we realized it did not support our point.

QGIS Digital Layout (March 8th)



This draft was mostly based on testing the planned word counts/text paragraphs on a projector to ensure the text was readable. On the advice of the professor, we decided to significantly pare-down the text on the actual map and include a 8.5x11 handout with additional text and our references/citations. At this stage we also removed the medium-scale inset as the original plan for overdose/neighbourhood statistics did not work out and we did not have a vision for an alternative metric to show.

Final Review Draft (April 5th)



Main Changes/Decisions (elaborated below)

- Re-added medium-scale map to show locations of other Ottawa ‘shroom’ stores
- Added LRT Train Lines to provide context to medium-scale map
- Designed a custom basemap using Mapbox studio to allow use of our colour palette and removal of unnecessary information.
- Changed positioning of textboxes to have information relevant to city councilor in the top left corner
- Decreased the amount of text on the page to make more readable
- Included logo/colour branding decisions

Feedback Received:

- Test ways in Mapbox Studio to make road labels less blurry
- Decrease saturation of buildings and Alta Vista ward to draw less attention
- Bold text of “City of Ottawa’s Community Safety …”
- Vertically centre bullet points text box in side panel
- Add drop shadow/frame and/or move textbox to side for light blue side panel

- Justify text in legislation box and fix the bolding of classification labels to make easier to read

Final Map (April 9th)

Design Decisions:

Scales/Map Extent:

When initially deciding on a location for our store, we worked with the Con map team to decide on a location that had a mix of positive and negative attributes for the creation of a 'shroom' store that was also in a different neighbourhood than the stores already in Ottawa. We decided on the Elmvale Acres Shopping Centre in Alta Vista since a wider view showed schools and the children's hospital (CHEO), as well as pharmacies, health services, Ottawa General, and the Perley senior and veteran services centre.

Due to this location the pro map decided on three scales to use in our map design. The first large-scale map would show only the immediate area around the store, making sure to avoid showing any schools and minimize the visibility of residential housing nearby. However we also used directional arrows and symbology to show positive features that are outside of the map frame, such as Ottawa General and Perley Health. Our medium scale map only shows the ward boundaries and locations of 'shroom' stores around the city, as well as contextual information such as the LRT lines to help the user identify the location of the store. Finally, the smallest scale map that we used showed the whole world in the WGS 84 projection as this is a familiar sight to many people inexperienced with cartography. One suggestion we received in

our final sprint review was to change the projection/small-scale map to not show the parts of Africa and Asia that do not have definitive laws on psychedelic use, however we decided to not pursue this to keep in line with typical expectations for maps and avoid explaining to the user why those countries were not visible.

Layout/Map Flow:

Our map layout was designed with the principles of map composition that were covered in Lecture 3 in mind. We used the focus of attention principle, which states that Western map readers will tend to move their eyes from the top left corner of the page, through the top middle and then down to the bottom. Due to this phenomena, we positioned the panel about City of Ottawa priorities in the top left corner, in order to open up the mind of the city councillor, before we mentioned the more controversial topic of magic mushrooms. After becoming more amenable to our message, the viewer's focus is then pulled to the location proposal in the large scale map, which also is the location with the more visual weight, being in the upper right of the page. Finally, the world map is located along the fringe of the viewer's perception in the bottom right of the map, as this is the least important/essential information and simply rounds out our message if the councillor still has reservations about the proposal. By structuring our map in this manner, we aim to ensure that we control the narrative of the map and do not immediately shut down our possibilities by opening with the terms "magic mushroom" or "psychedelic".

The other part of our map design relevant to the layout is the presence/amount of text on the map. As we wished to present a more 'facts forward' approach, our original design was very text heavy, however, after testing the design on a projector screen we realized it was very difficult to read some of our information, leading to the handout being designed as a supplement.

Branding/Colour Palette:

The colour design of our map was made to be reminiscent of the medical field, employing a blue/grey colour palette. This was chosen in contrast with the typical design of shroom stores as colourful, neon and focused on recreational use, in order to promote our store as an alternative to the city councillor.

Figure 1. Colour Palette generated with Coolors.co

These colours were used throughout our layout, handout, and maps to provide a unifying colour scheme and emphasise our goals and values. The colours were also given to Razz's sibling, who designed a more professional looking logo based on a sketch composed by Razz and Anna (Figure 2). This logo was used on the handout, as well as in the map to symbolize our store location and provide a contrasting point

feature to the other shroom stores, which were symbolized with a vibrant pink. For other features on the map, we used symbols with common and familiar meanings where relevant, such as a white “H” on a blue background for the hospital, and various coloured boxes with crosses on them to denote locations of medical services. The LRT lines were also denoted with red for the Confederation and green for the Trillium, which are the colours used by the city on planning and promotional documents.

Project Backlog:

Blue = Initial Backlog
Orange = Modified from Original
Green = Added Later
Red = Not Completed

Priority as denoted here refers to the order that things can/must be completed in, sequential priority assignments refer to tasks that rely on each other (ie. complete P3 before P4)

ID	Task Description	Point of Contact	Priority
1	Choose Location	Whole Group	1
2	Research Magic Mushrooms in Academic Literature	Razz	1
3	Research Legal Status of Psychedelics in Canada	Razz	1
4	Find data sources (neighborhood info, overdose/drug use statistics, nearby amenities)	Anna	2
5	Download and Cleanup data sources	Anna	3
6	Design Small Scale World Map of drug laws	Anna	4
~~7~~	~~Make graphs of other information related to magic mushroom use~~	~~NA~~	~~4~~
8	Write text for map handout	Razz	2
9	Design handout	Razz	3

10	Summarize handout text for map panels (Modified from Write Paragraphs x 3)	Razz	4
11	Identify background information for large scale map showing the location of the store (ie. OSM)	Anna	2
12	Design symbology on large scale map	Anna	4
13	Design Mapbox Studio Basemap	Anna	4
14	Layout and additional text information within large scale map	Razz	5
15	Prepare for Sprint Review Presentation(s)	Anna + Razz	NA
16	Weekly Team Sprint Meetings	Anna + Razz	NA
17	Final Review Presentation	Anna + Razz	6
18	Integrate Feedback + Produce Final Map	Anna	7
19	Data Management Plan	Razz	6
20	Compile Final Report	Razz	7

For the most part the project backlog was a faithful representation of the work we completed, as the scope of our map/project changed, such as the addition of an accompanying handout, some elements needed to be added (ID: 8,9), and others modified/combined due to text space limitations (ID: 10). Other elements of the backlog that were added referred to oversights from the initial project design phase, mainly related to the preparation for our final report (ID: 17,18,19,20). One element related to our cartography was added (ID: 13), as we originally planned to use a regular QGIS basemap for the project but realized after attempting that this clashed with the colour palette we designed and decided to instead use a Mapbox Studio custom basemap. Finally, we removed the inclusion/production of graphs within our map (ID: 7) as we did not find any data that was important enough to justify the valuable space it would have used.

Con Map

Scope:

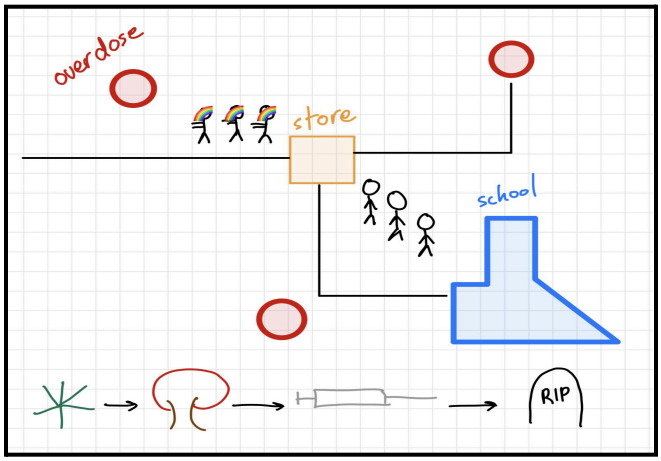
The purpose of this map is to demonstrate to the public in and around Alta Vista (Ward 18), the planned route for a protest against the opening of a magic mushroom (‘shroom’) store in the Elmvale Acres Shopping Centre, located at 1910 St.Laurent Boulevard, Ottawa, Ontario. This map will communicate the negative effects that the shop would have on the Alta Vista community, and aims to get the community involved in advocating for the closure of the shop. These effects include a potential misuse of the drug and increased usage. It is also illegal to sell or possess mushrooms, therefore this business would be operating illegally. This shop could also have an impact on kids since the proposed location is in close proximity to several schools and parks.

Design Description:

This map will be sized 8.5 x 11 as a handout that can be delivered door to door as well as posted at different gathering points within the neighbourhood and its neighbouring communities. The map will be designed as if it were created by a Facebook group of mom’s in the neighbourhood. The map will include a catchy hashtag as the title to help spread the intended message. It will also include graphics, the protest path and information, a QR code to a petition and social media handles for the neighbourhood group pages. The map will include a short text paragraph by the Alta Vista safety committee that states the concerns about the shop and details of the protest. Sources for the data will come from Open Ottawa and consist of roads, schools, hospitals, parks, overdose statistics.

Drafts:

Original Draft:



This draft represents the initial low-fidelity mockup that was quickly drawn to represent our general goals. The graphics in this draft were not included in any of the subsequent drafts. However the concept of showing overdoses in the community made it to the final map.

Draft #2:

Most of the aspects of this draft were kept and used in the final map. This draft introduced the concept of a protest in the community, as well as the general layout of our final map. The only thing that changed was that the drug pathway graphic in this draft was not used in the final map.

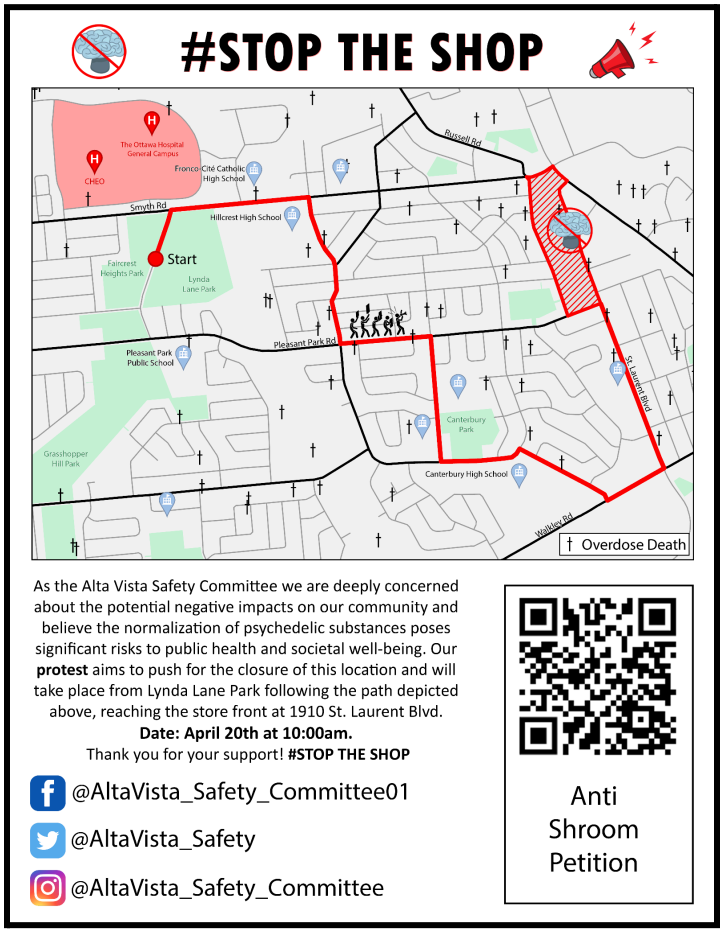
Map Shown in Presentation

This map built on the previous draft to include more detail. The symbology of the data was improved and the protest route was created. The text box was also written, expressing the community's concern about the shop as well as details of the protest. This draft is very similar to the final map. Presenting this draft gave us the opportunity to receive feedback, and make some adjustments to the final map.

Feedback on presented map:

- Overdose symbols more obvious
- Path of protest larger line
- Less side streets
- More obvious that hospital is CHEO
- More obvious this is a protest
- Imagine if you're seeing this poster on a street post

Final Map



From the previous map that was presented in class, several changes were made based on feedback provided.The overdose symbology was improved, in order to make it more visible. As well, the thickness of the protest pathway was increased, and the colour was changed for the less important streets, in order to make the protest route stand out. To make it more obvious that this map is about a protest, the word protest was bolded, and a graphic of people marking was included. The label of the hospital was also changed to make it more obvious that it is CHEO, the children's hospital.

Design Decisions:

Map Extent:

For our layout we used a large scale map only showing the neighbourhood surrounding the shop's proposed location. We chose to show the residential neighbourhood located behind the shop, as well as made sure that the nearby schools, parks, and childrens hospital were included in the map's extent.

Layout:

The layout of the map is designed to effectively convey the intended message of this map. The title is large and bolded at the top of the poster. For the title we chose to use a catchy hashtag to grab people's attention, and promote online recognition. The map of the protest route is intended to be the main focus of the layout. Therefore, it covers the full width of the page, so that it is easily visible. Below the map there is a smaller text box included, so that someone who is intrigued by the map can gain more information on what it is about. The QR code is located next to the text box, and is relatively large in size. This code was placed here, so that after the viewer reads the text if they wish to support the cause, they can easily sign the petition. Social media handles for the safety committee were then included at the bottom of the page. These were placed here so that, if the viewer is still interested, they can use these to gain more information, or to connect with the group.

Data choices/ Symbology:

For this map several layers were used to help convey the intended message. The protest route was included so that attendants are aware of the plan before the day of the protest, in order to reduce confusion. This route was symbolized as a thick red line so that it stands out from the roads, and attention is easily drawn to it. At the end of the route, the location for the store is symbolized as a polygon with red lines through it, as well as an anti-shroom graphic. This ensures that this is easily recognizable, and highlights the focus of the protest.

Nearby parks were displayed on the map as green areas.This data was included since drugs are commonly used in parks, and also because this is an area where children often are. Nearby schools were also included, and displayed as blue school symbols. This data was included to emphases the proximity of the store to many children. Likewise, the hospital was also included, and clearly labelled as CHEO, the

children's hospital. Including this hospital also emphases the close proximity of the shop to many children.

Data of overdoses was included to demonstrate the pre existing usage of drugs in the community. This was symbolized as crosses in order for it to stand out, since it is a distinctly different symbol from any other layers. For this data, dot density was used. This assigns a number of dots to random locations, however this information was purposely omitted from the map.The symbology implies that an overdose occured at every location of an overdose symbol.

Project Backlog:

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ID	Task Description	Effort Estimation (hours)	Priority
1	Research how magic mushroom stores can negatively affect a community (increased usage)	2	1
2	Neighbourhood statistics (~~how many children live in the area~~, schools, parks)	2	1
3	Download and Cleanup data sources	1	1
4	Plan out the layout of the handout	2	2
5	Write text paragraph on how to legally get shrooms	1	2
~~6~~	~~Create drug pathway graphic~~	~~1~~	~~2~~
7	Create **Protest** graphic	1	2

8	Add data and graphics to map, choosing an appropriate scale and projection	3	3
9	Manipulate data properties (labels, symbology)	2	3
10	Prepare for Final Presentation(s)	1/week	1
11	Integrate Feedback + Produce Final Map	1	1
12	Create data management plan	2	2
13	Final sprint report and final project map	1/week	1

Sprint Retrospective:

<u>Pro Map</u>

The final sprint of the project went very well. The pro group was able to create a high quality product to present to the Alta Vista City Councilor. Our group was able to communicate clearly to efficiently complete all remaining tasks of the project. As well, we were able to find lots of data that fit well with our scope and final map. We were also able to adapt if a data source did not work and quickly find a supplementary source. Our group was also able to create and agree upon design choices, such as a colour palette and symbology on the map, with no issues. One issue faced by the pro team during this sprint was with formatting. There were some issues with moving our draft in QGIS between a Mac and Windows computer, as well as with getting the text and labels to be properly positioned in the layout. These issues were fixed by experimenting with label placement, and adjusting parts of the layout lost between operating systems based on one another's layouts. There were also some technical issues when making the final edits to the map. Many elements would disappear upon export, which was solved by creating copies of each layout element and creating a new layout. The other problem faced during this sprint was with time management. Due to other classes, group members had less time to work on this sprint, which resulted in some stress when completing this sprint. This could be improved by adjusting the agile development process or using a different method of project management to better fit everyone's individual schedules.

Con Map

Overall, this sprint went well compared to previous sprints. Due to the longer time period between sprints, our group had ample time to meet up and discuss our project. We were able to determine exactly how we wanted the final layout to look, as well as make important symbology decisions for the final map. One struggle that our group faced was, determining which aspects of our previous drafts we wanted to include, and exclude. We had to determine what data, information, and graphics to include in order to effectively convey the intended message, however not overcrowd the map.

Both groups

Overall this project went well. Our group had a clear understanding of our objectives, and we were all on the same page regarding the project necessities. Our group faced some challenges when it came to communication especially between the pro and con teams. Each team worked very independently up until the final deadline, which led to some time management issues as the deadline approached. In the future, this project could be improved by ensuring our group has better communication, so that we can allow for better time management.